P(IK)

P(IK)[46]

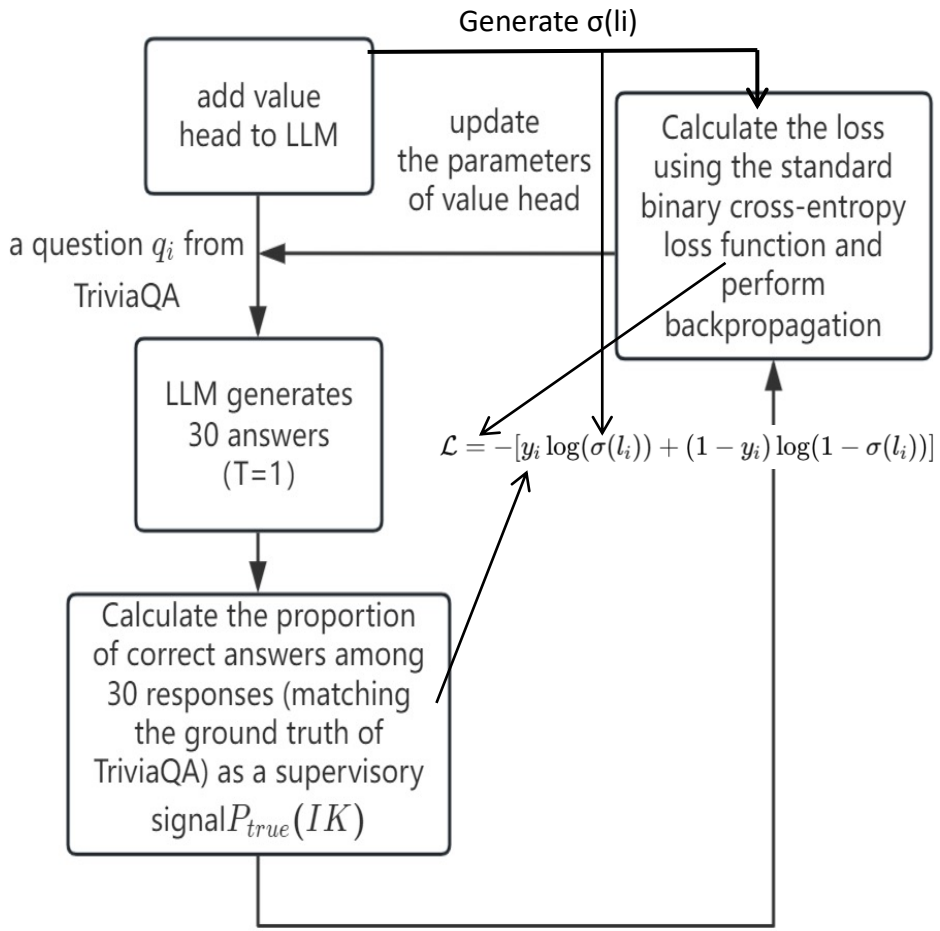
- **Task:** Estimates the probability that the model knows the correct answer before generating it.
- **Steps:**
  - Attach a scalar output layer to the transformer model.
  - This head outputs a single logit, converted via sigmoid:

$$\sigma(\text{logit}) = P(\text{IK})$$

  - Structure:

```
[Transformer Encoder/Decoder]
 ├── Language Modeling Head → predicts next token
 └── Value Head → outputs a scalar logit → σ(logit) = P(IK)
```

- **Train**



[46] Language Models (Mostly) Know What They Know, arxiv 2022

P(IK)[46]

- **Task Type:** Opening-end/Math/Code generation
- **Introduction:** A value head was added to the LLM to predict whether it "knows" the correct answer before generating it. The value head outputs a logit that is converted through a sigmoid function into the probability P(IK).

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Single-pass	-	White-box	Fine-tuning	High	Calibration/Ranking

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